B1 Laws of indices

Name:	
Class:	
Date:	

Time:	**217 minutes**
Marks:	**179 marks**
Comments:	

Q1.

Given that $\frac{dy}{dx} = \frac{1}{6x^2}$ find y

Circle your answer.

$\frac{-1}{3x^3} + c$ $\qquad$ $\frac{1}{2x^3} + c$ $\qquad$ $\frac{-1}{6x} + c$ $\qquad$ $\frac{-1}{3x} + c$

(Total 1 mark)

Q2.

$$f'(x) = \left(2x - \frac{3}{x}\right)^2 \text{ and } f(3) = 2$$

Find f(x).

(Total 4 marks)

Q3.

$$y = \frac{1}{x^2}$$

Find an expression for $\frac{dy}{dx}$

Circle your answer.

$\frac{dy}{dx} = \frac{0}{2x}$ $\qquad$ $\frac{dy}{dx} = x^{-2}$ $\qquad$ $\frac{dy}{dx} = -\frac{2}{x}$ $\qquad$ $\frac{dy}{dx} = -\frac{2}{x^3}$

(Total 1 mark)

Q4.

(a) Write down the value of p and the value of q given that:

(i) $\sqrt{3} = 3^p$

(1)

(ii) $\frac{1}{9} = 3^q$

(1)

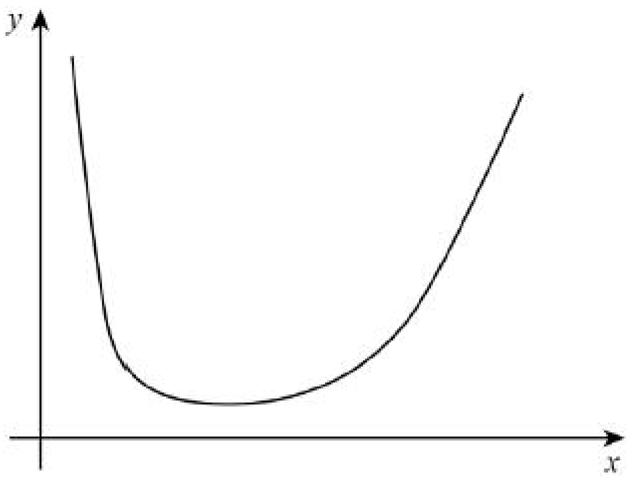
(b) Find the value of x for which $\sqrt{3} \times 3^x = \frac{1}{9}$

(2)

(Total 4 marks)

Q5.

A curve has equation $y = 6x^2 + \frac{8}{x^2}$ and is sketched below for $x > 0$



Find the area of the region bounded by the curve, the x-axis and the lines $x = a$ and $x = 2a$, where $a > 0$, giving your answer in terms of a

(Total 4 marks)

Q6.

A curve has equation $y = \frac{2}{\sqrt{x}}$

Find $\frac{dy}{dx}$

Circle your answer.

$\frac{\sqrt{x}}{3}$ $\qquad$ $\frac{1}{x\sqrt{x}}$ $\qquad$ $-\frac{1}{x\sqrt{x}}$ $\qquad$ $-\frac{1}{2x\sqrt{x}}$

(Total 1 mark)

Q7.

The height, x metres, of a column of water in a fountain display satisfies the differential equation $\frac{dx}{dt} = \frac{8 \sin 2t}{3\sqrt{x}}$, where t is the time in seconds after the display begins.

(a) Solve the differential equation, given that initially the column of water has zero height.
Express your answer in the form $x = f(t)$

(7)

(b) Find the maximum height of the column of water, giving your answer to the nearest cm.

(1)

(Total 8 marks)

Q8.

A circular ornamental garden pond, of radius 2 metres, has weed starting to grow and cover its surface.

As the weed grows, it covers an area of A square metres. A simple model assumes that the weed grows so that the rate of increase of its area is proportional to A.

(a) Show that the area covered by the weed can be modelled by

$$A = Be^{kt}$$

where B and k are constants and t is time in days since the weed was first noticed.

(4)

(b) When it was first noticed, the weed covered an area of 0.25 m^2. Twenty days later the weed covered an area of 0.5 m^2

(i) State the value of B.

(1)

(ii) Show that the the model for the area covered by the weed can be written as

$$A = 2^{\frac{t}{20} - 2}$$

(4)

(iii) How many days does it take for the weed to cover half of the surface of the pond?

(2)

(c) State one limitation of the model.

(1)

(d) Suggest one refinement that could be made to improve the model.

(1)

(Total 13 marks)

Q9.

A curve has the equation

$$y = \frac{12 + x^2\sqrt{x}}{x}, \quad x > 0$$

(a) Express $\frac{12 + x^2\sqrt{x}}{x}$ in the form $12x^p + x^q$.

(3)

(b) (i) Hence find $\frac{dy}{dx}$.

(2)

(ii) Find an equation of the normal to the curve at the point on the curve where $x = 4$.

(4)

(iii) The curve has a stationary point P. Show that the x-coordinate of P can be written in the form 2^k, where k is a rational number.

(3)

(Total 12 marks)

Q10.

(a) Write $\sqrt[4]{x^3}$ in the form x^k.

(1)

(b) Write $\dfrac{1-x^2}{\sqrt[4]{x^3}}$ in the form $x^p - x^q$.

(2)

(Total 3 marks)

Q11.

(a) Write down the values of p, q and r given that:

(i) $8 = 2^p$;

(1)

(ii) $\dfrac{1}{8} = 2^q$;

(1)

(iii) $\sqrt{2} = 2^r$.

(1)

(b) Find the value of x for which $\sqrt{2} \times 2^x = \dfrac{1}{8}$.

(2)

(Total 5 marks)

Q12.

(a) The expression $(2 + x^2)^3$ can be written in the form

$$8 + px^2 + qx^4 + x^6$$

Show that $p = 12$ and find the value of the integer q.

(3)

(b) (i) Hence find $\displaystyle\int \frac{(2+x^2)^3}{x^4}\,\mathrm{d}x$

(5)

(ii) Hence find the exact value of $\int_1^2 \frac{(2+x^2)^3}{x^4}\mathrm{d}x$.

(2)
(Total 10 marks)

Q13.

At the point (x, y) on a curve, where $x > 0$, the gradient is given by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 7\sqrt{x^5} - 4$$

(a) Write $\sqrt{x^5}$ in the form x^k, where k is a fraction.

(1)

(b) Find $\int\left(7\sqrt{x^5} - 4\right)\mathrm{d}x$.

(3)

(c) Hence find the equation of the curve, given that the curve passes through the point (1, 3).

(3)
(Total 7 marks)

Q14.

A curve C has the equation

$$y = \frac{x^3 + \sqrt{x}}{x}, x > 0$$

(a) Express $\frac{x^3 + \sqrt{x}}{x}$ in the form $x^p + x^q$.

(3)

(b) (i) Hence find $\frac{\mathrm{d}y}{\mathrm{d}x}$.

(2)

(ii) Find an equation of the normal to the curve C at the point on the curve where $x = 1$.

(4)

(c) (i) Find $\frac{\mathrm{d}^2y}{\mathrm{d}x^2}$.

(2)

(ii) Hence deduce that the curve C has no maximum points.

(2)
(Total 13 marks)

Q15.

(a) Write each of the following in the form $\log_a k$, where k is an integer:

(i) $\log_a 4 + \log_a 10$;

(1)

(ii) $\log_a 16 - \log_a 2$;

(1)

(iii) $3 \log_a 5$.

(1)

(b) Use logarithms to solve the equation $(1.5)^{3x} = 7.5$, giving your value of x to three decimal places.

(3)

(c) Given that $\log_2 p = m$ and $\log_8 q = n$, express pq in the form 2^y, where y is an expression in m and n.

(3)
(Total 9 marks)

Q16.

(a) Write down the value of n given that $\dfrac{1}{x^4} = x^n$.

(1)

(b) Expand $\left(1 + \dfrac{3}{x^2}\right)^2$.

(2)

(c) Hence find $\displaystyle\int \left(1 + \frac{3}{x^2}\right)^2 \, dx$.

(3)

(d) Hence find the exact value of $\displaystyle\int_1^3 \left(1 + \frac{3}{x^2}\right)^2 \, dx$.

(2)
(Total 8 marks)

Q17.

(a) (i) Find the value of p for which $\sqrt{125} = 5^p$.

(2)

(ii) Hence solve the equation $5^{2x} = \sqrt{125}$.

(1)

(b) Use logarithms to solve the equation $3^{2x-1} = 0.05$, giving your value of x to four decimal places.

(3)

(c) It is given that

$$\log_a x = 2(\log_a 3 + \log_a 2) - 1$$

Express x in terms of a, giving your answer in a form not involving logarithms.

(4)

(Total 10 marks)

Q18.

(a) Write $\sqrt{x^3}$ in the form x^k, where k is a fraction.

(1)

(b) A curve, defined for $x \geq 0$, has equation

$$y = x^2 - \sqrt{x^3}$$

(i) Find $\frac{dy}{dx}$.

(3)

(ii) Find the equation of the tangent to the curve at the point where $x = 4$, giving your answer in the form $y = mx + c$.

(5)

(Total 9 marks)

Q19.

A geometric series begins

$$20 + 16 + 12.8 + 10.24 + \ldots$$

(a) Find the common ratio of the series.

(1)

(b) Find the sum to infinity of the series.

(2)

(c) Find the sum of the first 20 terms of the series, giving your answer to three decimal places.

(2)

(d) Prove that the nth term of the series is 25×0.8^n.

(2)
(Total 7 marks)

Q20.

(a) Write down the values of p, q and r given that:

(i) $64 = 8^p$;

(ii) $\frac{1}{64} = 8^q$;

(iii) $\sqrt{8} = 8^r$.

(3)

(b) Find the value of x for which

$$\frac{8^x}{\sqrt{8}} = \frac{1}{64}$$

(2)
(Total 5 marks)

Q21.

(a) Simplify:

(i) $x^{\frac{3}{2}} \times x^{\frac{1}{2}}$;

(1)

(ii) $x^{\frac{3}{2}} \div x$;

(1)

(iii) $\left(x^{\frac{3}{2}}\right)^2$

(1)

(b) (i) Find $\int 3x^{\frac{1}{2}}\,dx$.

(3)

(ii) Hence find the value of $\int_1^9 3x^{\frac{1}{2}}\,dx$.

(2)

(Total 8 marks)

Q22.

(a) Write $\sqrt{x}$ in the form x^k, where k is a fraction.

(1)

(b) Hence express $\sqrt{x}\,(x-1)$ in the form $x^p - x^q$.

(2)

(c) Find $\int \sqrt{x}(x-1)\,dx$.

(3)

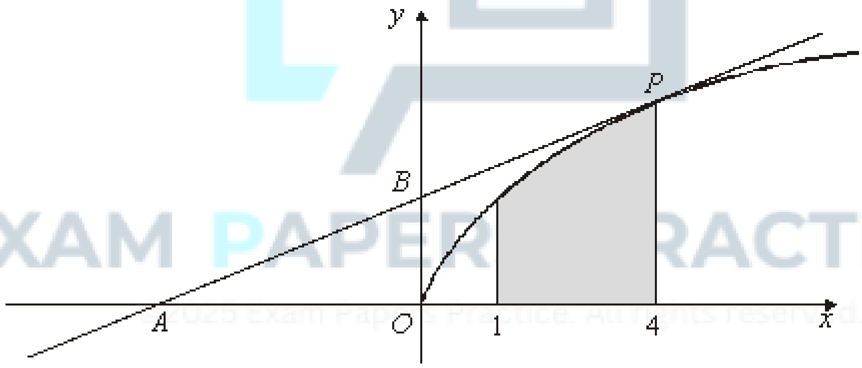
(d) Hence show that $\int_1^2 \sqrt{x}(x-1)\,\mathrm{d}x = \frac{4}{15}\left(\sqrt{2}+1\right)$.

(3)

(Total 9 marks)

Q23.

The diagram shows a curve C with equation $y =$. The point O is the origin (0, 0).



The region bounded by the curve C, the x-axis and the vertical lines x = 1 and x = 4 is shown shaded in the diagram.

(a) (i) Write $\sqrt{x}$ in the form x^p, where p is a constant.

(1)

(ii) Find $\int \sqrt{x}\,\mathrm{d}x$.

(2)

(iii) Hence find the area of the shaded region.

(3)

(b) The point on C for which x = 4 is P. The tangent to C at the point P intersects the x-axis and the y-axis at the points A and B respectively.

(i) Find an equation for the tangent to the curve C at the point P.

(4)

(ii) Find the area of the triangle AOB.

(3)

(c) Describe the single geometrical transformation by which the curve with equation $y = \sqrt{x-1}$ can be obtained from the curve C.

(2)

(d) Use the trapezium rule with four ordinates (three strips) to find an approximation for $\int_1^4 \sqrt{x-1}\,dx$, giving your answer to three significant figures.

(4)

(Total 19 marks)

Q24.

A curve is defined, for $x > 0$, by the equation $y = f(x)$, where

$$f(x) = \frac{x^8 - 1}{x^3}$$

(a) Express $\frac{x^8 - 1}{x^3}$ in the form $x^p - x^q$, where p and q are integers.

(2)

(b) (i) Hence differentiate $f(x)$ to find $f'(x)$.

(2)

(ii) Hence show that f is an increasing function.

(2)

(c) Find the gradient of the normal to the curve at the point (1, 0).

(3)

(Total 9 marks)